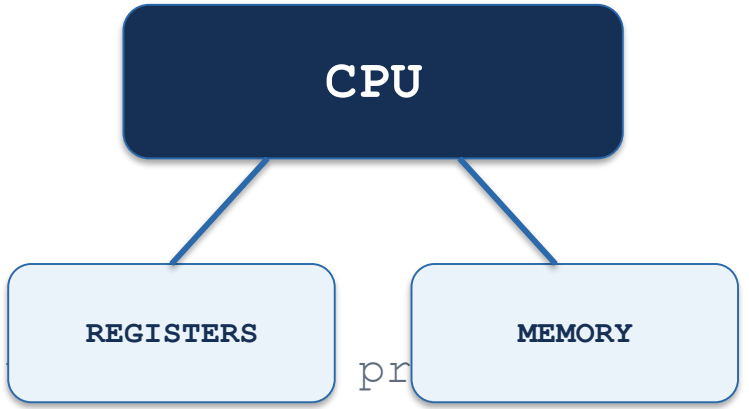
The RISC-V Processor



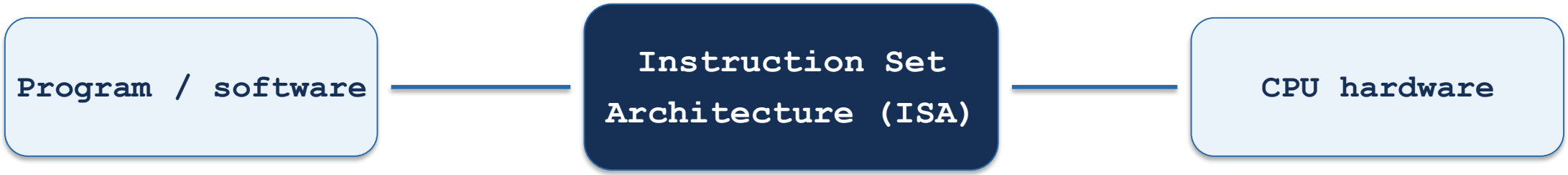
How instructions, datapath and control work together to pr

RV32I | Instruction set | Datapath | Control unit

Concepts → architecture → working computer

1. What is an instruction set architecture?

ISA is the agreement between software and the processor.



RISC-V

An open instruction set architecture used f

RV32I

A base RISC-V instruction set with 32-bit inte

2. What instructions can do

Most beginner examples fit into a few useful groups.

Arithmetic / logic

ADD, SUB, AND, OR

Calculate or compare values.

Memory access

LW, SW

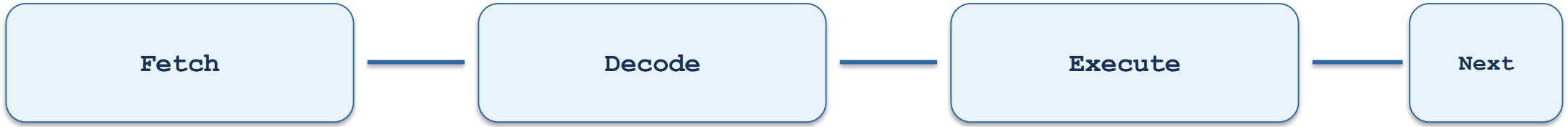
Load a value from memory or st

Control flow

BEQ, JAL

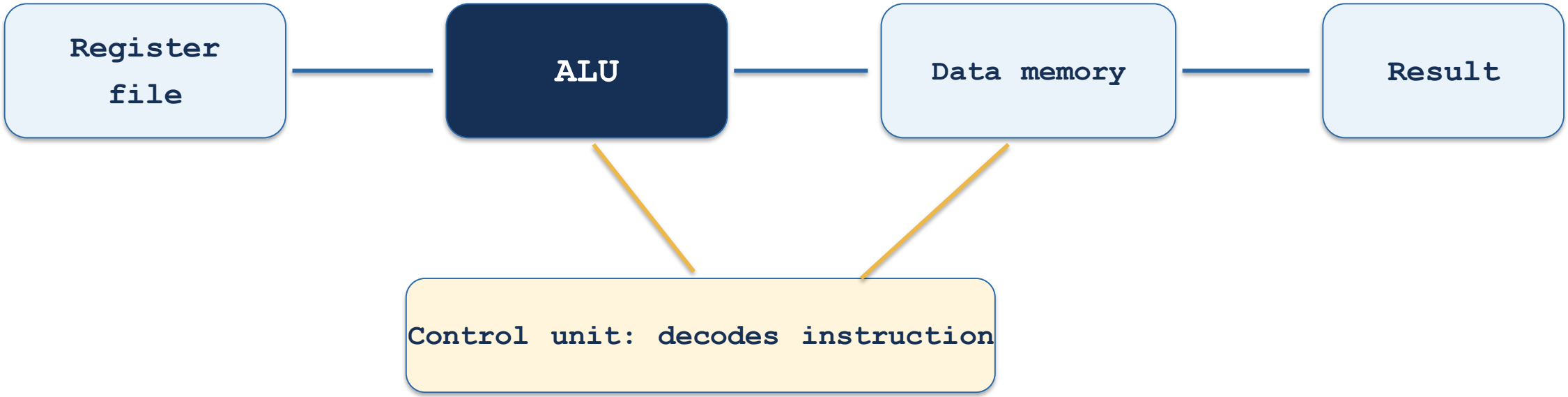
Choose the next instruction or jump

Example idea: an instruction names an operation and the registers or values it uses.



3. Inside the processor: datapath + control

The datapath moves and processes values; control tells it what to do.



The control signals select register inputs, ALU operation, memory actions and where the resu

4. A single-cycle CPU

A simple model completes one instruction in one clock cycle.



Why build this first?

It makes the full instruction path visible a

Trade-off

A single-cycle design is simple, but the clock must

5. From assembly to hardware

An instruction passes through the CPU as encoded bits.

Example assembly instruction



Meaning: add the values in x1 and x2, then store the result in x3.

Assembler translates human-readable assembly into machine code. The CPU decodes that code in